

Monitor active compliance tasks

- Ensures compliance activities continue to be tracked after task generation.
- Retrieves all active (OPEN and IN_PROGRESS) tasks from MongoDB during scheduled monitoring runs.

Detect approaching and overdue deadlines

- Enables proactive compliance by identifying tasks that require immediate attention.
- Compares task due dates with the current date and classifies tasks as on-track, approaching deadline, or overdue based on configurable thresholds.

Generate recurring task instances

- Recurring regulatory obligations must continue without manual intervention.
- Creates the next task occurrence using the configured recurrence rule while maintaining links to previous occurrences.

Detect compliance gaps

- Identifies operational failures beyond simple overdue tasks.
- Detects approved obligations without tasks, stale tasks without deadlines, orphaned obligations, and other compliance gaps, storing them in the `compliance_gaps` collection.

Update task status

- Keeps the compliance lifecycle synchronized with real-world task progress.
- Updates task statuses (e.g., OPEN → OVERDUE), records monitoring timestamps, and persists status changes in MongoDB.

Maintain tamper-evident audit log

- Every compliance action must be traceable and protected against tampering.
- Uses the centralized Audit Log Service to append hash-chained audit entries for task updates, recurring task generation, compliance gaps, and monitoring activities.

Record monitoring run metadata

- Provides visibility into monitoring execution and supports operational troubleshooting.
- Creates `monitoring_runs` records containing execution time, scanned tasks, overdue tasks, generated recurrences, detected gaps, and run status.

Update compliance metrics

- Compliance teams require accurate, real-time operational insights.
- Aggregates task statistics, overdue counts, recurrence information, and compliance gap metrics for dashboard reporting.

Provide audit history and dashboard data

- Enables complete traceability and real-time compliance visibility.
- Makes audit records, monitoring results, task summaries, and compliance metrics available to the Dashboard and Traceability views.

Receive pending obligations

- Ensures only AI-extracted obligations are presented for human validation before operational use.
- Retrieves pending obligation records from MongoDB using the obligation_ids received from the Obligation Extraction Agent.

Assemble prioritized review queue

- Enables reviewers to focus on the most critical or uncertain obligations first.
- Sorts obligations based on confidence score, ambiguity flags, and review status to create a prioritized review queue.

Present clause and AI extraction to reviewer

- Allows reviewers to compare the original regulatory clause with the AI-generated obligation.
- Displays the clause text, extracted obligation, entity, deadline, frequency, evidence type, confidence score, and ambiguity flags in the review interface.

Capture approve, edit, or reject decisions

- Human validation ensures regulatory obligations are accurate before becoming compliance tasks.
- Records reviewer actions including approval, edits, rejection reason, reviewer identity, and review timestamp.

Validate reviewer edits

- Ensures modified obligation data remains complete and structurally valid.
- Validates edited fields using predefined schemas and business rules before saving the updated obligation.

Preserve original AI extraction

- Maintains transparency and traceability between the original AI output and the final reviewed obligation.
- Stores the original extraction separately while recording reviewer modifications without overwriting the AI-generated data.

Record reviewer attribution

- Every compliance decision must be attributable to an authorized reviewer.
- Stores reviewer details, review timestamps, decision history, and comments for complete auditability.

Update obligation status

- Reflects the current lifecycle state of each obligation for downstream processing.
- Updates obligation status to APPROVED, EDITED, or REJECTED and persists the changes in MongoDB.

Forward approved obligations to the Task Generation Agent

- Ensures only validated obligations proceed to operational task generation.
- Updates the ExecutionContext with approved obligation_ids and review metadata before forwarding them to the Task Generation Agent.

Receive clause IDs

- Ensures only validated legal clauses are processed for obligation extraction.
- Retrieves `clause_ids` and `segmentation_run_id` from the `ExecutionContext` and loads clause documents from MongoDB.

Retrieve clause information

- Complete legal context is required for accurate obligation extraction.
- Fetches clause text, hierarchy, section details, parent clause, and metadata from MongoDB.

Pre-screen candidate clauses

- Reduces unnecessary AI processing by filtering non-actionable clauses.
- Uses deterministic keyword and mandatory language detection (e.g., shall, must, required, maintain, submit) to identify obligation candidates.

Batch candidate clauses

- Improves extraction efficiency while avoiding API rate limits.
- Groups candidate clauses into token-aware batches and instructs the LLM to process each clause independently.

Extract structured obligations

- Regulatory obligations require semantic understanding beyond rule-based processing.
- Uses a structured LLM prompt to extract obligation details including entity, deadline, frequency, evidence type, confidence, and ambiguity information.

Validate extracted output

- Ensures only valid structured data enters the compliance pipeline.
- Validates the LLM response using Pydantic models and retries failed extractions before marking them for manual review.

Calculate confidence score

- Helps reviewers prioritize uncertain AI extractions.
- Combines LLM confidence with deterministic validation heuristics to produce a composite confidence score.

Flag ambiguities

- Makes uncertain or incomplete extractions transparent to reviewers.
- Identifies issues such as missing deadlines, cross-references, compound clauses, and possible hallucinations.

Save obligation records to DB

- Creates structured compliance objects for downstream processing.
- Stores obligation documents with PENDING_REVIEW status and records extraction metadata in extraction_runs.

Forward obligation IDs

- Enables the next agent to continue processing without duplicating data.
- Updates the ExecutionContext with obligation_ids and extraction_run_id before forwarding to the Human Review Agent.

Receive normalized document

- Ensures segmentation always starts from a clean, standardized document produced by the Ingestion Agent.
- Receives document_id and retrieves normalized text and metadata from MongoDB.

Detect document hierarchy

- Regulatory documents are organized into chapters, sections, clauses, and sub-clauses that must be preserved.
- Uses regex and numbering patterns (e.g., 1, 1.1, (a), (i)) to identify hierarchy levels.

Identify clause boundaries

- Obligations exist at clause boundaries, not arbitrary token boundaries.
- Splits text using legal numbering and sentence boundaries instead of fixed-size chunks.

Preserve legal hierarchy

- Parent-child relationships provide context for downstream obligation extraction and traceability.
- Stores parent_clause_id, hierarchy_path, and section information with every clause.

Generate clause records

- Every clause becomes an independently addressable unit for later processing.
- Creates one MongoDB document per clause containing clause number, text, metadata, and hierarchy.

Validate segmentation

- Detects structural issues before AI processing begins.
- Checks numbering sequence, orphan clauses, duplicate numbering, and malformed sections.

Save clause information to DB

- Makes segmented clauses available to downstream agents.
- Persists clause documents and creates a segmentation_runs record.

Forward clause IDs

- Downstream agents only need references, not duplicate data.
- Updates the ExecutionContext with clause_ids and segmentation_run_id.

Receive approved obligations

- Ensures only human-approved obligations are converted into compliance tasks.
- Retrieves approved obligation records from MongoDB using the obligation_ids received from the Human Review Agent.

Check for existing tasks (idempotency)

- Prevents duplicate task creation caused by retries or duplicate approval events.
- Checks whether a task already exists for the obligation_id before generating a new task.

Map obligation fields to task fields

- Converts validated regulatory obligations into structured operational tasks.
- Deterministically maps obligation details such as title, description, evidence type, and compliance metadata to the task object without using an LLM.

Resolve default owner assignment

- Every compliance task requires a responsible owner for accountability.
- Resolves the task owner using configurable owner-mapping rules based on applicable entity, evidence type, and obligation context, with a default fallback owner when no rule matches.

Compute deadlines and recurrence

- Compliance tasks require actionable deadlines and recurring obligations must continue automatically.
- Computes due dates, recurrence rules, deadline type, and identifies tasks requiring manual deadline assignment when deadlines cannot be derived.

Assign task priority

- Helps compliance teams focus on higher-risk regulatory activities first.
- Assigns task priority using deterministic business rules based on penalty references, deadline characteristics, and obligation metadata.

Create task records

- Every approved obligation should become an independently trackable compliance task.
- Creates a task document containing owner, deadline, recurrence, evidence requirements, priority, status, and traceability information.

Store task generation run

- Maintains complete traceability of the task generation process.
- Stores task documents, links them to their originating obligations, and records execution details in the task_generation_runs collection.

Forward tasks to the Continuous Monitoring & Audit Agent

- Enables continuous monitoring of generated compliance tasks throughout their lifecycle.
- Updates the ExecutionContext with task_ids and task_generation_run_id before forwarding them to the Continuous Monitoring & Audit Agent.

